Status	Finished
Started	Sunday, 11 October 2026, 4:50 PM
Completed	Sunday, 11 October 2026, 4:56 PM
Duration	5 mins 46 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction

b. Selection

c. Distribution

d. Insertion

The correct answer is:
Insertion

Question **2**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It only works on small datasets

b. It simplifies the sorting process by using only one pass

c. It avoids the need for recursion

d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **3**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It has no effect on the search operation
c. It increases the number of comparisons needed
d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **4**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sorted()
b. sort()
c. order()
d. arrange()

The correct answer is: sorted()

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Insertion Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Makes data harder to manage
c. Makes code execution slower
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are combined to form a single sorted list
b. They are compared element by element
c. They are discarded
d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Dividing the input into smaller subproblems
c. Comparing each element with the others
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **9**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Inserting data into a list
- b. Searching for data in a list
- c. Arranging data in a particular format
- d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **10**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. arrange()
- c. order()
- d. sorted()

The correct answer is: sort()

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It sorts data out of the box efficiently
- b. It is less efficient than custom sorting algorithms
- c. It only works with integer arrays
- d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **12**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It makes the analysis slower
- c. It complicates the analysis process
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **13**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort cannot handle duplicate elements
- b. Bubble Sort is less efficient for large lists
- c. Bubble Sort does not guarantee sorted order
- d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting is rarely used in programming
- b. Sorting decreases the efficiency of selection operations
- c. Sorting helps in finding duplicates quickly
- d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Merge Sort
c. Bubble Sort
d. Binary Search

The correct answer is: Bubble Sort